Ganesh Gupta

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Experience

Oracle | *Software Engineer, AI Platform* Jun 2025 – Present

- Owned full-stack delivery of backend services and APIs (Python/FastAPI, React/TypeScript) handling millions of daily transactions at 99%+ uptime.
- Built a reusable **distributed executor** platform exposing typed **REST APIs** at **sub-200ms p99**.
- Replaced polling with SSE for live metrics streaming, cutting redundant API calls 90% through performance tuning and architectural changes.
- Participated in design and code reviews enforcing secure coding standards and testability, and triaged production issues affecting service reliability.

QureAI | *ML Infrastructure Engineer* Mar 2026 – Present

- Built a model observability framework tracking drift, latency, and eval regressions in production, enabling governance-aligned rollback workflows.
- Cut p99 inference latency 420ms to 140ms across 50K+ daily requests by migrating synchronous serving to async on GCP Cloud Run with autoscaling.
- Raised **top-3 retrieval recall from 61% to 89%** on a 2K-query eval set by rebuilding the pipeline with **hybrid dense-sparse retrieval** and **re-ranking**.

Nomura Research Institute Financial Technologies | *Software Engineer* Sep 2019 – Aug 2023

- Architected 25+ production Java Spring Boot microservices exposing REST APIs for financial transaction processing for 1M+ users at 99.9% uptime on AWS (EC2, RDS, SQS).
- Cut nightly financial data aggregation blocking downstream settlement from 8 minutes to 90 seconds by redesigning SQL with indexed views and batch partitioning.
- Standardized REST API failure-handling contracts across 25+ Java microservices, reducing escalations 60% via consistent retry, timeout, and OAuth-aligned authentication patterns.
- Led **architecture and code reviews** and mentored junior engineers on system design.

Projects

Watch-your-LLM `github.com/ganeshhgupta/watch-your-llm`

- Full-stack platform with FastAPI backend, React/TypeScript dashboard, and PostgreSQL, exposing REST APIs for rubric-based scoring and latency tracking at scale.

Distributed KV Store / VPC Control Plane Simulator `github.com/ganeshhgupta/distributed-kv-store`

- Raft-consensus distributed KV store handling concurrent writes with linearizable reads and log compaction; companion VPC control plane simulator modeling subnet routing.

RootCause AI `github.com/ganeshhgupta/rootcause-ai`

- Multi-agent system automating production incident diagnosis via tool-use chains; routes alerts to specialized sub-agents and synthesizes root cause reports.

Education

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0* Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science* Jul 2016 – Jul 2020

Skills

Languages: Python, Java, SQL, TypeScript

Networking and Distributed Systems: Subnet Routing, Traffic Forwarding, Raft Consensus, Microservices, REST APIs, Kafka, SQS, Fault Tolerance, Low-Latency Services

Performance and Ops: Latency Optimization, Profiling, Data Structures and Algorithms, Observability, Prometheus/Grafana, AWS, GCP, Docker, Kubernetes, CI/CD

Certifications: AWS Certified Developer Associate, GCP Professional MLOps Engineer